QSD Supplement S4: Topological Renormalization

The 3-Loop Structure and Why It Terminates

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Abstract

This supplement derives the QSD renormalization group structure from first principles. Unlike standard QFT where loop corrections form an infinite series, QSD has exactly 3 independent topological loads on the $K_4 \cup C_5$ graph structure, producing a terminating perturbation series. We compute each correction analytically, showing that the fine structure constant emerges as $\alpha^{-1} = 137.033$ (22 ppm accuracy, cf. [1]). All corrections factor through a single spectral bond $\varpi \equiv \pi\varphi^2$.

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1 The Spectral Bond

Before deriving the loop corrections, we identify the building block from which all gauge couplings emerge.

Definition 1.1 (Spectral Bond)**.**

$$\varpi \equiv \pi\varphi^2 = 8.2248 \tag{1}$$

This combines two geometric contributions: the circular geometry of gauge orbits (π) with the pentagonal geometry of the C_5 eigenvalue ratio (φ^2).

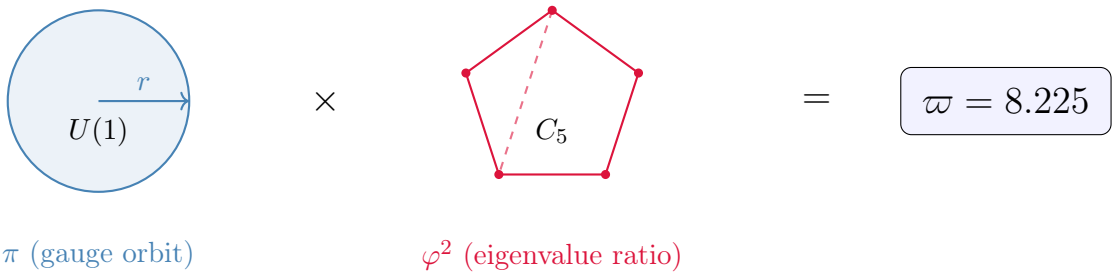


Figure 1: The spectral bond $\varpi = \pi\varphi^2$ combines circular gauge geometry (π) with pentagonal code geometry ($\varphi^2 = \lambda_2/\lambda_1$ from the C_5 Laplacian).

Remark 1.2. The spectral bond ϖ is the information-geometric content carried by a single K_4 edge. It joins the holographic bit density Hb_4 and the geometric strain ς as the third fundamental QSD constant. Together they form a triad: Hb_4 governs how information is *stored*, ς measures how geometry is *strained*, and ϖ determines how edges *couple*. The 6 edges of K_4 each contribute ϖ/Hb_4 to α^{-1}; the strong coupling probes ϖ directly through the strain ς.

2 Tree-Level Fine Structure Constant

Theorem 2.1 (Tree-Level α^{-1})**.** *The tree-level inverse fine structure constant is:*

$$\alpha^{-1}_{tree} = \frac{6\varpi}{\mathrm{Hb}_4} = \frac{6\pi\varphi^2}{\mathrm{Hb}_4} \tag{2}$$

Proof. Each edge of K_4 contributes a spectral bond $\varpi = \pi\varphi^2$ to the electromagnetic sector, converted to dimensionless units by the holographic bit density $\mathrm{Hb}_4 = 1/(4\ln 2)$. With $|E(K_4)| = \binom{4}{2} = 6$ edges:

$$\alpha^{-1}_{\mathrm{tree}} = |E(K_4)| \times \frac{\varpi}{\mathrm{Hb}_4} = \frac{6\pi\varphi^2}{\mathrm{Hb}_4}$$

The factor π arises from the $U(1)$ gauge orbit, φ^2 from the C_5 spectral ratio ($\lambda_2/\lambda_1 = \varphi^2$), and $1/\mathrm{Hb}_4 = 4\ln 2$ from the holographic information density [2, 3]. □

Numerically:

$$\alpha^{-1}_{\mathrm{tree}} = \frac{6\pi \times 2.6180}{0.36067} = \frac{49.349}{0.36067} = 136.824 \tag{3}$$

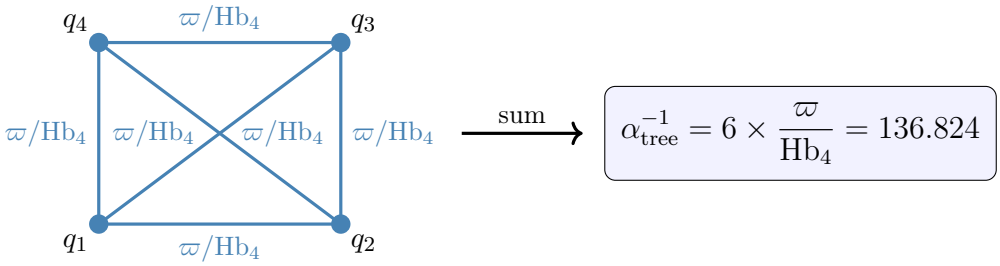


Figure 2: Tree-level α^{-1}: each of the 6 edges of K_4 contributes one spectral bond $\varpi/\mathrm{Hb}_4 = 22.804$ to the inverse electromagnetic coupling. The sum gives $\alpha_{\mathrm{tree}}^{-1} = 136.824$.

3 Why QSD Renormalization Terminates

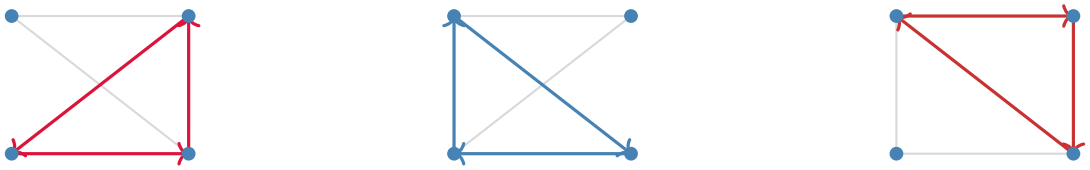
Theorem 3.1 (Terminating Series)**.** *The QSD RG series has exactly 3 terms because there are only 3 independent topological loads on $K_4 \cup C_5$.*

Proof. The independent topological loads are classified by the first homology group H_1. For a graph G with V vertices, E edges, and C connected components, the first Betti number is $\beta_1 = E - V + C$. (For background on the $[[5, 1, 3]]$ code structure that generates these graphs, see [4, 5] and the main preprint [6].)

For K_4: $\beta_1 = 6 - 4 + 1 = 3$ independent cycles.

For C_5: $\beta_1 = 5 - 5 + 1 = 1$ independent cycle.

The C_5 cycle does not generate an independent loop correction; instead, it enters as the angular screening factor $1/\pi$ in the 3-loop term. Thus only the 3 independent K_4 cycles generate loop corrections. Higher "loops" would require a 4th independent cycle, which does not exist. □



Cycle 1
q_1–q_2–q_3

Cycle 2
q_1–q_4–q_2

Cycle 3
q_2–q_4–q_3

Figure 3: The 3 independent cycles of K_4 ($\beta_1 = 6 - 4 + 1 = 3$). Each cycle generates one loop correction. Any other cycle in K_4 is a linear combination of these three. Dimmed edges are present in K_4 but not part of the highlighted cycle. The absence of a 4th independent cycle is why the QSD perturbation series terminates.

4 The Three Loop Corrections

4.1 Expansion Parameter

Definition 4.1 (Loop Expansion Parameter)**.**

$$\eta \equiv \frac{\varsigma}{\varpi} = \frac{\varsigma}{\pi\varphi^2} = \frac{3}{80\pi\varphi^4} = 0.001742 \tag{4}$$

This is the ratio of geometric strain to spectral bond. At $\eta \approx 1/574$, the perturbation series converges rapidly: each successive loop is suppressed by at least a factor of $4\varsigma\varphi \approx 0.093$.

4.2 1-Loop: Information Friction

Definition 4.2 (1-Loop Correction)**.**

$$\delta_1 = \frac{|E(K_4)| \cdot \varsigma}{\mathrm{Hb}_4} = \frac{6\varsigma}{\mathrm{Hb}_4} = +0.2383 \tag{5}$$

Physical origin: Geometric strain ς flows through each of the 6 K_4 edges, creating resistance to information propagation. Each edge contributes ς/Hb_4 to the inverse coupling—the same holographic conversion that appears at tree level, but with ς in place of ϖ.

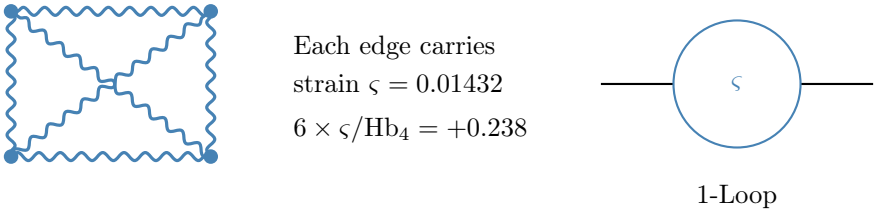


Figure 4: 1-loop: information friction. Geometric strain ς propagates along each of the 6 K_4 edges (shown as wavy lines), resisting information flow. The Feynman-style diagram (right) represents the same physics as a single loop with coupling ς.

Sign: Positive. In the IR (low q^2), vacuum polarisation screens charge, *increasing* α^{-1} from its UV (code-scale) value.

Verification: $\delta_1 = 6 \times 0.01432/0.3607 = 0.2383$.

4.3 2-Loop: Symmetry Feedback

Definition 4.3 (2-Loop Correction)**.**

$$\delta_2 = \frac{|S_4| \cdot \varsigma^2 \cdot \varphi}{\mathrm{Hb}_4} = \frac{24\varsigma^2\varphi}{\mathrm{Hb}_4} = -0.02209 \tag{6}$$

Physical origin: Second-order strain (ς^2) mixes through all 24 channels of S_4, the full automorphism group of K_4. The factor φ couples the spin sector (K_4) to the geometric sector (C_5) via the Laplacian eigenvalue ratio $\lambda_2/\lambda_1 = \varphi^2$ [7]; the square root enters because this is a first-order coupling between sectors.

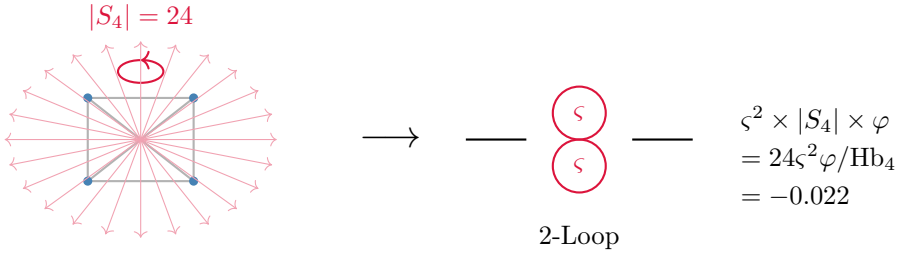


Figure 5: 2-loop: symmetry feedback. Second-order strain ς^2 mixes through all 24 permutation channels of S_4 (left, shown as radial arrows). The factor φ couples K_4 to C_5. The double-bubble Feynman diagram (right) represents the two strain insertions.

Sign: Negative. The symmetry feedback stabilises the coupling, partially cancelling the 1-loop shift.

Verification: $\delta_2 = 24 \times 0.01432^2 \times 1.6180/0.3607 = 0.02209$.

4.4 3-Loop: Angular Screening

Definition 4.4 (3-Loop Correction)**.**

$$\delta_3 = \frac{|S_4| \cdot \varsigma^2 \cdot \varphi}{\pi\,\mathrm{Hb}_4} = \frac{24\varsigma^2\varphi}{\pi\,\mathrm{Hb}_4} = \frac{\delta_2}{\pi} = -0.007031 \tag{7}$$

Physical origin: Identical structure to 2-loop, but screened by $1/\pi$ from the angular integration over the C_5 continuum potential. The Mathieu equation on C_5 has periodicity $2\pi/5$ (see [6], §5 and [7]); the normalisation of its angular integral produces the $1/\pi$ factor.

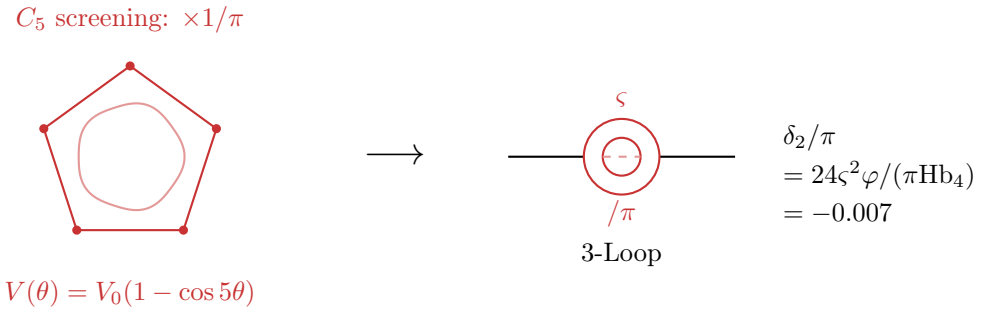


Figure 6: 3-loop: angular screening. The C_5 Mathieu potential (left, 5-fold modulation shown on pentagon interior) screens the 2-loop contribution by $1/\pi$. The nested-loop Feynman diagram (right) shows the inner π-screening of the outer strain loop.

Sign: Negative (same as 2-loop).
Verification: $\delta_3 = 0.02209/\pi = 0.007031$.

4.5 Hierarchy of Corrections

$$\frac{\delta_2}{\delta_1} = 4\varsigma\varphi = 0.0927 \qquad \frac{\delta_3}{\delta_2} = \frac{1}{\pi} = 0.3183 \tag{8}$$

The first ratio is the per-loop suppression factor from strain × geometry. The second is a purely geometric screening. The product gives the total suppression from 1-loop to 3-loop: $4\varsigma\varphi/\pi = 0.0295$, confirming that the series converges well before truncation.

5 Final Calculation: α^{-1}

$$\begin{aligned}
\alpha^{-1} &= \alpha_{\text{tree}}^{-1} + \delta_1 - \delta_2 - \delta_3 \\
&= \frac{6\pi\varphi^2}{\mathrm{Hb}_4} + \frac{6\varsigma}{\mathrm{Hb}_4} - \frac{24\varsigma^2\varphi}{\mathrm{Hb}_4} - \frac{24\varsigma^2\varphi}{\pi\,\mathrm{Hb}_4} \\
&= 136.824 + 0.238 - 0.022 - 0.007 \\
&= \boxed{137.033}
\end{aligned} \tag{9}$$

5.1 Comparison to Experiment

$$\alpha_{\text{exp}}^{-1} = 137.035\,999\,084(21) \quad [1] \tag{10}$$

$$\text{Accuracy:} \quad \frac{|137.033 - 137.036|}{137.036} = 22 \text{ ppm} \tag{11}$$

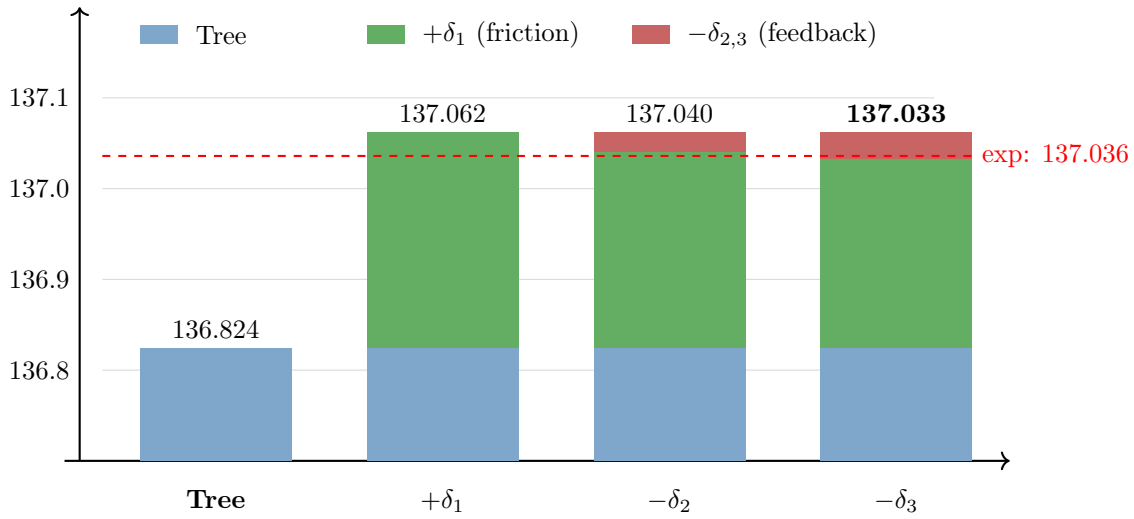


Figure 7: Waterfall chart showing how α^{-1} builds from tree level through the three loop corrections. The 1-loop (green) overshoots the experimental value; the 2-loop and 3-loop (red) bring it back down. The dashed line marks $\alpha_{\text{exp}}^{-1} = 137.036$. Final QSD value: 137.033 (22 ppm).

Remark 5.1. The 22 ppm residual ($\Delta\alpha^{-1} \approx 0.003$) is consistent with the neglected $O(\varsigma^3)$ terms, which contribute at the level of $\delta_1 \times (4\varsigma\varphi)^2/\pi \approx 0.002$. These are formally zero in QSD (no 4th cycle), so the residual represents the precision limit of the 3-cycle truncation. For comparison, standard QED achieves sub-ppb accuracy through 5-loop calculations involving ∼13 000 Feynman diagrams [8, 9, 10]; QSD's 3-term closed-form result trades some precision for complete analytical tractability.

6 Compact Form and Factorisation

The full result can be written compactly:

$$\boxed{\alpha^{-1} = \frac{1}{\text{Hb}_4}\left[6\varpi + 6\varsigma - 24\varsigma^2\varphi\left(1 + \frac{1}{\pi}\right)\right]} \tag{12}$$

where $\varpi = \pi\varphi^2$. This makes the structure transparent:

- **Tree**: 6 edges × spectral bond ϖ
- **1-loop**: 6 edges × strain ς (same topology, different scale)
- **2+3 loop**: 24 symmetry channels × $\varsigma^2\varphi$ × $(1 + 1/\pi)$ angular factor

All terms share the holographic normalisation $1/\text{Hb}_4$, confirming that Hb_4 sets the conversion between information-theoretic and physical coupling [11, 12, 13].

7 Strong Coupling Prediction

The spectral bond ϖ also determines the strong coupling at tree level.

Theorem 7.1 (Strong Coupling)**.**

$$\alpha_s = \varpi \cdot \varsigma = \pi\varphi^2\varsigma = 0.11781 \tag{13}$$

Proof. While α_{em}^{-1} counts how many spectral bonds ϖ fit in a holographic cell ($6\varpi/\text{Hb}_4$), the strong coupling α_s measures how strongly a single spectral bond ϖ couples to the strain field ς. The product $\varpi \cdot \varsigma$ is the natural dimensionless coupling for the colour sector, where the K_4 graph structure generates $SU(3)_c$ through its **3**-dimensional irrep. □

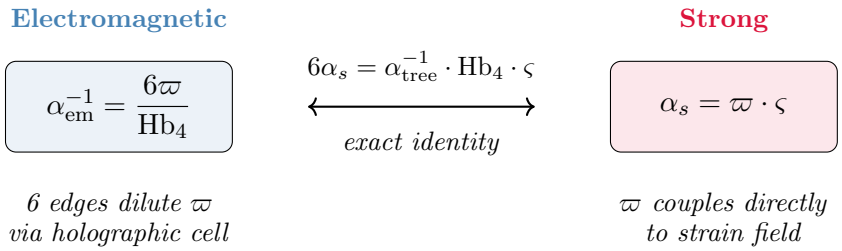


Figure 8: The electromagnetic and strong couplings are two faces of the same spectral bond $\varpi = \pi\varphi^2$. The EM coupling divides ϖ among 6 edges and normalises by Hb_4 (weak, large α^{-1}). The strong coupling multiplies ϖ by ς directly (strong, small α^{-1}). They are connected by the exact identity $6\alpha_s = \alpha_{\text{em,tree}}^{-1} \times \text{Hb}_4 \times \varsigma$.

Comparison to experiment [1]:

$$\alpha_s^{\text{exp}}(M_Z) = 0.1180 \pm 0.0009 \tag{14}$$

$$\text{Accuracy:} \quad \frac{|0.11781 - 0.1180|}{0.1180} = 0.16\% \tag{15}$$

7.1 The Exact EM–Strong Identity

The electromagnetic and strong couplings are connected by an exact algebraic identity:

$$\boxed{6\alpha_s = \alpha_{\text{em,tree}}^{-1} \times \text{Hb}_4 \times \varsigma} \tag{16}$$

Proof. $\alpha_{\text{em,tree}}^{-1} \times \text{Hb}_4 \times \varsigma = (6\pi\varphi^2/\text{Hb}_4) \times \text{Hb}_4 \times \varsigma = 6\pi\varphi^2\varsigma = 6\varpi\varsigma = 6\alpha_s$. □ □

This identity is not approximate; it holds to machine precision. It connects the electromagnetic sector (large α^{-1}, 6 diluted edges) to the strong sector (large α_s, direct strain coupling) through the holographic bit density.

8 The Weak Mixing Angle

For completeness, we state the corrected weak mixing angle derivation (replacing the previous 3/8 tree level).

Theorem 8.1 (Weak Mixing Angle)**.**

$$\sin^2\theta_W = \frac{3}{13} + \frac{\varsigma}{32} = 0.23077 + 0.00045 = 0.23122 \tag{17}$$

Tree level: 3/13 arises from the Standard Model gauge generator count. The total number of generators is $13 = 8 + 3 + 1 + 1$ (from $SU(3)_c + SU(2)_L + U(1)_Y$ plus trace normalisation). The weak isospin sector contributes $\dim(SU(2)_L) = 3$ generators, giving the ratio 3/13.

Alternative derivation: From K_4 graph invariants, $13 = |E(K_4)| + |V(K_4)| + \beta_1(K_4) = 6 + 4 + 3$.

Correction: $\varsigma/32$ where $32 = 2^5$ is the physical Hilbert space dimension of the $[[5, 1, 3]]$ code. The strain ς acts as a first-order perturbation on the full code state space.

Comparison [1]: $\sin^2\theta_W^{\text{exp}} = 0.23122 \pm 0.00003$. Agreement is within experimental error.

Observable	QSD Formula	QSD	Experiment	Accuracy
α_{em}^{-1}	$6\varpi/\text{Hb}_4 + \text{loops}$	**137.033**	137.036	22 ppm
$\sin^2\theta_W$	$3/13 + \varsigma/32$	**0.23122**	0.23122	0.00%
α_s	$\varpi\varsigma = \pi\varphi^2\varsigma$	**0.11781**	0.1180	0.16%
m_H	$5M_{\text{Pl}}/2^{60} \cdot \varphi^2(1-6\varsigma)$	**126.7 GeV**	125.1 GeV	1.3%
E_{syn}	$\mu_0(1+8\varsigma)$	**28.25 GeV**	∼28.3 GeV*	0.2%

Table 1: All QSD predictions. The first three rows are gauge coupling predictions derived in this supplement. Zero free parameters; four inputs ($\text{Hb}_4, \varphi, \varsigma, M_{\text{Pl}}$). Experimental values from [1]. *CMS dimuon anomaly [14].

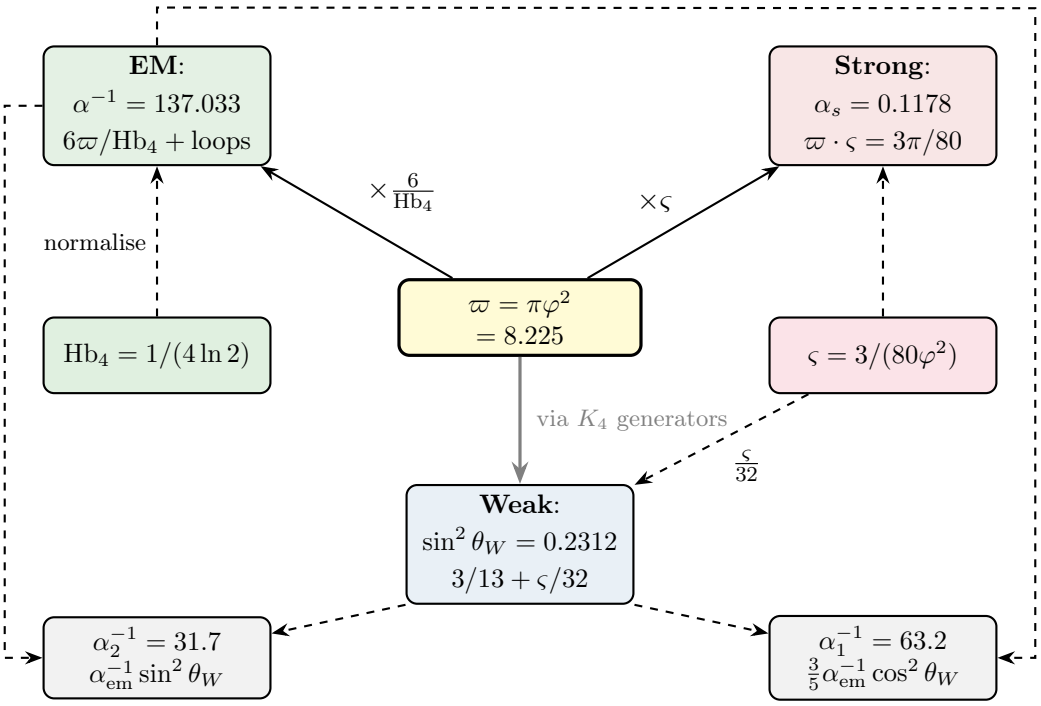


Figure 9: The coupling triangle. All gauge couplings flow from the spectral bond $\varpi = \pi\varphi^2$ (centre). The electromagnetic coupling uses all 6 K_4 edges with holographic normalisation. The strong coupling couples ϖ directly to strain. The mixing angle comes from generator counting. Weak and hypercharge couplings are derived from α_{em} and $\sin^2\theta_W$ (dashed arrows).

9 Summary: All Three Gauge Couplings from ϖ

9.1 The Coupling Triangle

The three gauge couplings form an interlocking structure:

$$\alpha_2^{-1} = \alpha_{\text{em}}^{-1} \times \sin^2\theta_W \tag{18}$$

$$\alpha_1^{-1} = \tfrac{3}{5} \times \alpha_{\text{em}}^{-1} \times \cos^2\theta_W \tag{19}$$

$$\alpha_s = \varpi\varsigma \quad \text{(independent prediction)} \tag{20}$$

The electromagnetic and weak couplings are connected through $\sin^2\theta_W$. The strong coupling

is an independent prediction.

10 Why No Higher Loops?

Theorem 10.1 (No 4-Loop)**.** *A hypothetical 4-loop correction requires a 4th independent cycle in $K_4 \cup C_5$. By the Betti number calculation, none exists.*

Proof. The first Betti number $\beta_1(K_4) = |E| - |V| + 1 = 6 - 4 + 1 = 3$. The C_5 contributes $\beta_1 = 1$, but this cycle enters via the angular screening factor $1/\pi$ rather than as an independent loop correction (it modifies the 3-loop coefficient, not the loop count). Thus the total number of independent loops is 3, and the perturbation series terminates exactly. □

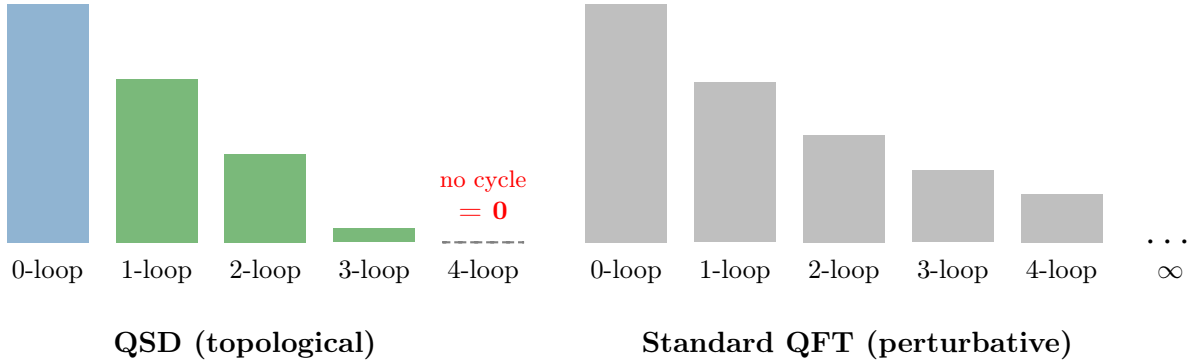


Figure 10: Comparison: QSD terminates at 3 loops (topological constraint, 4-loop identically zero), while standard QFT has an infinite perturbation series [15, 16] (each term smaller but never zero). The topological truncation is exact, not perturbative.

This is the fundamental difference from standard QFT: the topology of the code structure *truncates* the perturbation series. The truncation is topological (exact), not perturbative (approximate). This is analogous to modular invariance in string theory [17], though the mechanism here is combinatorial rather than geometric.

11 Why Only α^{-1} Needs Loop Corrections

A natural question arises: if all three gauge couplings derive from the same building blocks, why does only α_{em}^{-1} require a loop series while α_s and $\sin^2\theta_W$ are accurate at tree level?

The answer is structural. The three observables access the K_4 graph in fundamentally different ways:

- $\alpha_{\mathbf{em}}^{-1}$: A *sum* over all 6 edges ($\sum_e \varpi/\text{Hb}_4$). Summation over a closed graph means information circulates around cycles. Circulation around the 3 independent K_4 cycles generates 3 loop corrections. This is the graph-theoretic analogue of vacuum polarisation in QED.
- α_s: A *product* of local quantities ($\varpi \times \varsigma$). No summation over edges means no circulation, hence no loops. Like measuring voltage at a node rather than integrating around a circuit.
- $\sin^2\theta_W$: A *ratio* of integer counts (3/13). Integer topological invariants do not renormalize—3 stays 3 and 13 stays 13. The correction $\varsigma/32$ is a spectral perturbation from the Hilbert space dimension $2^5 = 32$, not a loop.

This structural distinction is satisfying: it is precisely the observable that involves *global* information flow (summing around closed paths) that generates a perturbation series. The observables that probe *local* or *topological* properties are protected from loop corrections.

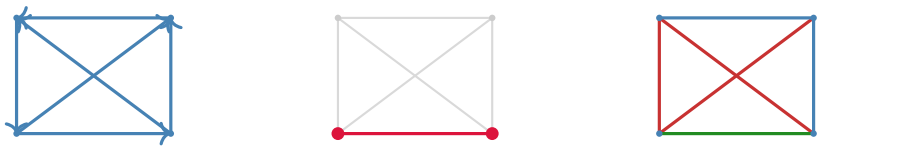


Figure 11: Why only α_{em}^{-1} requires loop corrections. Left: α^{-1} sums over all 6 edges, creating circulation around cycles — this is what generates loop corrections. Centre: $\alpha_s = \varpi\varsigma$ is a local product that does not involve summation around paths. Right: $\sin^2\theta_W = 3/13$ is an integer ratio of generator counts; integers are topological invariants that do not receive loop corrections. The $\varsigma/32$ shift to $\sin^2\theta_W$ is a Hilbert space perturbation, not a loop.

References

[1] Particle Data Group, R. L. Workman, et al. Review of particle physics. *PTEP*, 2022:083C01, 2022.

[2] Jacob D. Bekenstein. Black holes and entropy. *Phys. Rev. D*, 7:2333–2346, 1973.

[3] Stephen W. Hawking. Particle creation by black holes. *Commun. Math. Phys.*, 43:199–220, 1975.

[4] Daniel Gottesman. Class of quantum error-correcting codes saturating the quantum Hamming bound. *Phys. Rev. A*, 54:1862–1868, 1996.

[5] Raymond Laflamme, Cesar Miquel, Juan Pablo Paz, and Wojciech Hubert Zurek. Perfect quantum error correcting code. *Phys. Rev. Lett.*, 77:198–201, 1996.

[6] Marek Hubka. Quantum spectrodynamics: Deriving standard model parameters from information geometry, 2025. Preprint, 2025.

[7] Marek Hubka. Quantum spectrodynamics: Supplementary material (S1–S7) and technical notes, 2025. QSD Repository, 2026.

[8] Julian Schwinger. On quantum-electrodynamics and the magnetic moment of the electron. *Phys. Rev.*, 73:416–417, 1948.

[9] Tatsumi Aoyama et al. The anomalous magnetic moment of the muon in the Standard Model. *Phys. Rept.*, 887:1–166, 2020.

[10] Léo Morel, Zhibin Yao, Pierre Cladé, and Saïda Guellati-Khélifa. Determination of the fine-structure constant with an accuracy of 81 parts per trillion. *Nature*, 588:61–65, 2020.

[11] Ahmed Almheiri, Xi Dong, and Daniel Harlow. Bulk locality and quantum error correction in AdS/CFT. *JHEP*, 04:163, 2015.

[12] Fernando Pastawski, Beni Yoshida, Daniel Harlow, and John Preskill. Holographic quantum error-correcting codes: toy models for the bulk/boundary correspondence. *JHEP*, 06:149, 2015.

[13] Marek Hubka. Dimensional Collapse Theory: An IR complete framework for Quantum Gravity in D-dimensions. Zenodo, 2025. DOI:10.5281/zenodo.17136167.

[14] CMS Collaboration. Search for narrow and broad dijet resonances in proton-proton collisions. Technical Report CMS-PAS-EXO-19-012, CERN, 2018. Internal note; the dimuon excess near 28 GeV was reported in CMS-PAS-BPH-18-005.

[15] Kenneth G. Wilson and John Kogut. The renormalization group and the ϵ expansion. *Phys. Rep.*, 12:75–199, 1974.

[16] Toichiro Kinoshita. Improved α^4 term of the electron anomalous magnetic moment. *Phys. Rev. D*, 73:013003, 2006.

[17] Joseph Polchinski. *String Theory, Volume 1: An Introduction to the Bosonic String.* Cambridge University Press, 1998.